

Instructor: Date:

Learner: Regional Branch:

Program: Center:

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Let's Start!

Pronunciation Spot: Would you clone yourself if you could?



Learning Outcome

Describe events in the past through a wider range of B1 language in both work and social contexts.

Ask your partner:

- What do you know about cloning?
- What questions do the cloning of animals and humans bring to mind?

Can Do Statements

- I can say and use words like *technical*.
- I can say and pronounce questions like "How long have you been ... ?"
- I can stress new information.

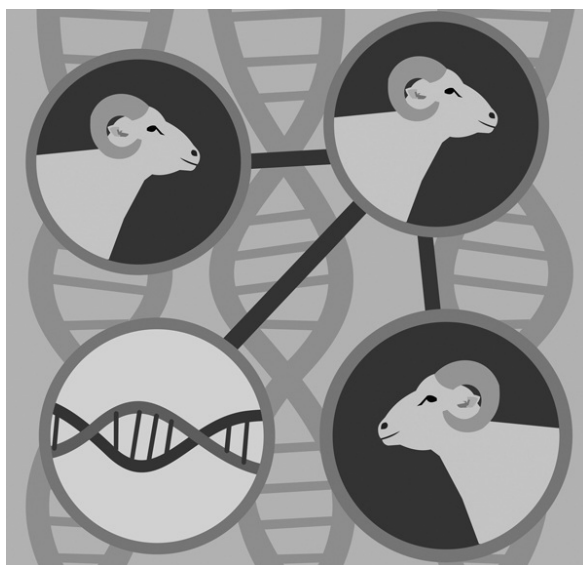




Let's Explore!

Genetics studies how **characteristics*** are passed on to offspring.

* The "*characteristics*" or traits that they study are physical ones.



Instructions: Read the text aloud.

What do you know about genetic engineering? Genetics is a field of science. It studies how characteristics are passed on from parent to offspring. Genetic engineering investigates how genetics can be used and applied.

The best known area of genetic engineering is cloning. How can you describe cloning? Cloning produces a group of living things genetically identical to the parent. It's already a practical natural reproductive solution for some plants and animals. Why do scientists want to experiment with artificial cloning? What have they done so far in this area? In 1997, Scottish scientists made headlines when they successfully cloned a sheep using genetic material from a six-year-old ewe.

Researchers saw cloning endangered animals as the logical next step. Others were critical, viewing cloning as radical, immoral and impractical - the stuff of science fiction. Would you someday keep a personal clone? How would you feel when you used those clones for replacement parts?

In reality, how far along have we actually come with cloning? To date, the failure rate with animals, which was supposed to be very low, is still very high. Plant cloning, on the other hand, is an economical way to produce uniform gardens with a consistent, equal growth rate.





Let's Practice!

Sandra reviews her notes.

Instructions: Organize the mystery words. Read the words aloud. Take into account the pronunciation of ical. Check your pronunciation with your instructor.

1. nelciaitd _____
2. atraicpcl _____
3. rltai fcia _____
4. aaltmire _____
5. gliaolc _____
6. liactrci _____



Let's Practice!

John has some questions about cloning.

Instructions: Match the two parts of the sentence from column A to column B.

COLUMN A

1. What do you know
2. Why do scientists want
3. How can you
4. How far along have

COLUMN B

- a. we come with cloning?
- b. to experiment with artificial cloning?
- c. about genetic engineering?
- d. describe cloning?





Extension Activity

Discuss the pros and cons of cloning.

Instructions: Complete the following steps. Fill in the Self-Assessment.

- Consider what it would mean if everyone could have a medical clone.
- Make a list of pros and cons of this practice.
- Would they be human? Property? Where would they live?
- What would the benefits be?
- Where would they live?
- Who would bring them up?
- Present your list of pros and cons to another student.
- Listen to their ideas and discuss them all.

Self-Assessment

Yes

No

I list the pros and cons of cloning humans.

☐ ☐

I present my list to a partner, paying careful attention to pronunciation.

☐ ☐

I discuss my ideas and my partner's ideas.

☐ ☐


Target Words

- **to clone** (verb): to make multiple identical copies of (a DNA sequence)
e.g. Genetic engineers clone a group of living things so they could be genetically identical to the parent.
- **offspring** (noun): the immediate descendant or descendants of a person, animal, etc.; progeny
e.g. Her offspring have become world famous.
- **genetic material** (noun): the genetic composition of living organisms
e.g. Scottish scientists made headlines when they successfully cloned a sheep using genetic material from a six-year-old ewe.
- **critical** (adjective): inclined to find fault or to judge with severity, often too readily
e.g. Some researchers were critical of cloning.
- **radical** (adjective): a very different perspective from the conventional or traditional point of view
e.g. Some researchers viewed cloning as radical.
- **impractical** (adjective): not practical or useful
e.g. Cloning is viewed as an impractical procedure.
- **immoral** (adjective): contrary to established moral principles
e.g. Some researchers were critical, viewing cloning as radical, immoral and impractical - the stuff of science fiction.
- **economical** (adjective): designed or functioning to make effective use of money or effort invested
e.g. Plant cloning is an economical way to produce uniform gardens.





Learning Points

Words Ending in Suffix *-ical*

The suffix *-ical* is used to transform nouns to adjectives. It means "of or pertaining to".

Some of the examples of words ending in *-ical* are: *mythical, chemical, alphabetical and technical.*

You pronounce the suffix *-ical* as "icul"

Practice saying these words out loud:

identical
practical
artificial
material
logical
critical
radical
impractical
economical

Examples:

- *Cloning produces a group of living things genetically identical to the parent.*
- *It's already a practical natural reproductive solution for some plants and animals.*
- *Why do scientists want to experiment with artificial cloning?*
- *In 1997, Scottish scientists made headlines when they successfully cloned a sheep using genetic material from a six-year-old ewe.*
- *Researchers saw cloning endangered animals as the logical next step.*
- *Others were critical, viewing cloning as radical, immoral and impractical.*
- *Plant cloning, on the other hand, is an economical way to produce uniform gardens.*





Learning Points

Reduction of Auxiliary Verbs and Pronouns in Questions

When you speak, it is common to use the reduced form of modals and auxiliaries in sentences and questions.

Reduction is very common in conversations. It doesn't happen when you write. When you use a reduced form of auxiliary verbs in questions, you link the words together when you say them.

Examples:

What do you mean? = **Whatchu** mean?

What do you know about genetic engineering? = **Whatchu** know about genetic engineering?

How would you feel when you used those clones for replacement parts? = **Howdjya** feel when you used those clones for replacement parts?

When you use pronouns other than *you*, the pronoun itself is not reduced, but is still part of the 3-word glide.

Examples:

What have they done so far in this area? = **Whaduvthey** done so far in this area?

In reality, how far have we actually come with cloning? = In reality, how **faruvwe** actually come with cloning?

